

COMPREHENSIVE EXAM SYSTEM

Application, Presentation, Session, Transport,
Network, Data Link & Physical Layers

ISO/OSI Model & TCP/IP Protocol Suite

Lecture 3-1 — Dr. Mohammad Adly

Full Question Bank • Answers • Page References • Final Exam • Answer Key

Total Questions	Question Types	Pages Covered	Final Exam
250+	MCQ, T/F, Fill, Short, Match	Pages 1-83	50 Hard Questions

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SECTION 1: APPLICATION LAYER

Pages 1-3 | Lecture by Dr. Mohammad Adly

Q1: The Application Layer serves as a _____ to lower layers.

Type: MCQ

- A) Bridge
- B) User interface
- C) Router
- D) Physical connector

Correct Answer: B

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Q2: Which of the following is a function of the Application Layer?

Type: MCQ

- A) Routing packets
- B) Bit stream transmission
- C) Preparation of data for each service
- D) Frame error checking

Correct Answer: C

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Q3: The Application Layer is responsible for end-to-end communication on both sides.

Type: True / False

Correct Answer: True

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Q4: Hardware devices reside in the Application Layer.

Type: True / False

Correct Answer: False

Page: Page 2

Note: Software applications reside in it, not hardware.

Q5: The Application Layer sends _____ and receives _____.

Type: Fill in the Blank

Correct Answer: requests / reply

Page: Page 2

Q6: List three main functions of the Application Layer.

Type: Short Answer

Correct Answer: User interface to lower layers, Preparation of data for each service, End-to-end communication on both sides.

Page: Page 2

Q7: Which protocol is used for transferring web pages?

Type: MCQ

- A) FTP
- B) SMTP
- C) HTTP
- D) POP3

Correct Answer: C

Page: Page 3

Q8: Which protocol is used for sending emails?

Type: MCQ

- A) HTTP
- B) SMTP
- C) DNS
- D) RTP

Correct Answer: B

Page: Page 3

Q9: SSH stands for:

Type: MCQ

- A) Simple Shell
- B) Secure Shell
- C) Session Shell
- D) Standard Shell

Correct Answer: B

Page: Page 3

Q10: Which protocol resolves domain names to IP addresses?

Type: MCQ

- A) SMTP
- B) FTP
- C) DNS
- D) RTP

Correct Answer: C

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Q11: RTP stands for Real-time Protocol.

Type: True / False

Correct Answer: True

Page: Page 3

Q12: POP3 stands for Post Office Protocol version 3.

Type: True / False

Correct Answer: True

Page: Page 3

Q13: HTTP stands for _____.

Type: Fill in the Blank

Correct Answer: Hyper Text Transfer Protocol

Page: Page 3

Q14: FTP stands for _____.

Type: Fill in the Blank

Correct Answer: File Transfer Protocol

Page: Page 3

Q15: IMAP stands for _____.

Type: Fill in the Blank

Correct Answer: Internet Message Access Protocol

Page: Page 3

Q16: Which of the following is NOT an Application Layer protocol?

Type: MCQ

A) HTTP

B) FTP

C) ARP

D) SMTP

Correct Answer: C

Page: Page 3

Note: ARP works between Network and Data Link layers.

Q17: Match each abbreviation with its full name:

Type: Matching

1) HTTP

2) FTP

3) SMTP

4) DNS

Match with:

a) File Transfer Protocol

b) Domain Name Server

c) Simple Mail Transfer Protocol

d) Hyper Text Transfer Protocol

Correct Answer: 1-d, 2-a, 3-c, 4-b

Page: Page 3

Q18: List all Application Layer protocols mentioned in the lecture.

Type: Short Answer

Correct Answer: HTTP, FTP, SMTP, POP3, IMAP, Telnet/SSH, DNS, RTP

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Q19: Telnet and SSH are both Application Layer protocols.

Type: True / False

Correct Answer: True

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Q20: POP3 stands for:

Type: MCQ

A) Post Office Protocol version 3

B) Post Office transfers Protocol

C) Primary Office Protocol

D) Packet Office Protocol

Correct Answer: A

Page: Page 3

SECTION 2: PRESENTATION LAYER

Page 4

Q21: The Presentation Layer is responsible for:

Type: MCQ

- A) End-to-end routing
- B) Common representation of data between source and destination
- C) Physical addressing
- D) Session management

Correct Answer: B

Page: Page 4

Q22: The Presentation Layer provides transformation of data.

Type: True / False

Correct Answer: True

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Q23: Data encryption is a function of the Transport Layer.

Type: True / False

Correct Answer: False

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Note: Data encryption/decryption is a function of the Presentation Layer.

Q24: The Presentation Layer handles data _____ / decryption and data _____ / decompression.

Type: Fill in the Blank

Correct Answer: encryption / compression

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Q25: List all functions of the Presentation Layer.

Type: Short Answer

Correct Answer: Common representation of data, transformation of data, standardized application interface, coding of data syntax, data encryption/decryption, data compression/decompression.

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Q26: Which of the following is NOT a function of the Presentation Layer?

Type: MCQ

- A) Data compression
- B) Data encryption

- C) Coding of data syntax
- D) Flow control

Correct Answer: D

Page: Page 4

Note: Flow control is a function of the Transport Layer.

Q27: The Presentation Layer supports standardized application interface.

Type: True / False

Correct Answer: True

Page: Page 4

Q28: Coding of _____ is a function of the Presentation Layer.

Type: Fill in the Blank

Correct Answer: data syntax

Page: Page 4

Q29: Data decompression is handled by which layer?

Type: MCQ

- A) Application
- B) Presentation
- C) Session
- D) Transport

Correct Answer: B

Page: Page 4

Q30: The Presentation Layer ensures common data representation between source and destination.

Type: True / False

Correct Answer: True

Page: Page 4

SECTION 3: SESSION LAYER

Pages 5-8

Q31: Which of the following is a function of the Session Layer?

Type: MCQ

- A) Data encryption
- B) Establishing sessions
- C) Routing
- D) Frame checking

Correct Answer: B

Page: Page 5

Q32: List the four main session operations handled by the Session Layer.

Type: Short Answer

Correct Answer: Establishing Session, Managing Session, Controlling Session, Terminating Session.

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Q33: The Session Layer is responsible for dialogue synchronization.

Type: True / False

Correct Answer: True

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Q34: The Session Layer cannot reconnect if a connection cuts off.

Type: True / False

Correct Answer: False

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Note: The Session Layer reconnects if a connection cuts off.

Q35: The Session Layer defines node communication _____ and _____.

Type: Fill in the Blank

Correct Answer: order / timing

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Q36: Which layer is responsible for checking for connection failure and reconnecting?

Type: MCQ

- A) Application
- B) Session
- C) Transport
- D) Network

Correct Answer: B

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Q37: What additional functions does the Session Layer provide beyond basic session management?

Type: Short Answer

Correct Answer: Making sure the connection is active during session, securing the connection, dialogue synchronization, checking for connection failure, reconnecting if connection cuts off, terminating communication, defining node communication order and timing.

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Q38: The Application, Presentation, and Session layers are closely connected and related in function.

Type: True / False

Correct Answer: True

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Q39: In TCP/IP, the Application, Presentation, and Session layers are merged to form the _____ layer.

Type: Fill in the Blank

Correct Answer: Application

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Q40: The top three OSI layers (Application, Presentation, Session) in TCP/IP form the:

Type: MCQ

- A) Transport layer
- B) Network layer
- C) Application layer
- D) Physical layer

Correct Answer: C

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Q41: Content formatting and preparing for network process are roles of the upper three OSI layers combined.

Type: True / False

Correct Answer: True

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Q42: Securing the connection is a function of which OSI layer?

Type: MCQ

- A) Physical
- B) Data Link
- C) Session
- D) Network

Correct Answer: C

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SECTION 4: TRANSPORT LAYER

Pages 9-15 & 56-65

Q43: The Transport Layer organizes data into:

Type: MCQ

- A) Packets
- B) Frames
- C) Segments
- D) Bits

Correct Answer: C

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Q44: List the main functions of the Transport Layer.

Type: Short Answer

Correct Answer: Organize data into segments, reliable end-to-end transport, loss recovery, flow control, data ordering, data deduplication, best effort delivery.

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Q45: The Transport Layer provides loss recovery.

Type: True / False

Correct Answer: True

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Q46: The Transport Layer provides either reliable transport or _____ delivery.

Type: Fill in the Blank

Correct Answer: Best Effort

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Q47: Data Deduplication is a function of which layer?

Type: MCQ

- A) Network
- B) Transport
- C) Data Link
- D) Session

Correct Answer: B

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Q48: MTU stands for:

Type: MCQ

- A) Maximum Transfer Utility

- B) Maximum Transmission Unit
- C) Minimum Transmission Unit
- D) Maximum Transport Utility

Correct Answer: B

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Q49: What is Segmentation in the Transport Layer?

Type: Short Answer

Correct Answer: Dividing data into small fragments that can be transmitted over the network. Different networks have different MTU sizes.

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Q50: MTU defines the maximum size of one data piece carried through the network.

Type: True / False

Correct Answer: True

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Q51: MTU acknowledgment happens during _____ routine and while _____ connection with the recipient.

Type: Fill in the Blank

Correct Answer: discovering / establishing

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Q52: Reassembly of data segments is done at:

Type: MCQ

- A) The sending end
- B) The receiving end
- C) Both ends
- D) The router

Correct Answer: B

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Q53: Data can be received in different order and the Transport Layer rearranges it.

Type: True / False

Correct Answer: True

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Q54: What is the purpose of Sequencing in the Transport Layer?

Type: Short Answer

Correct Answer: Using sequencing identifiers to avoid losses and duplication, fix the issue of data following different paths, organize data in groups, and identify the order of data sending and interpretation.

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Q55: Sequencing avoids losses and duplication.

Type: True / False

Correct Answer: True

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Q56: Sequencing uses _____ identifiers to organize data segments.

Type: Fill in the Blank

Correct Answer: sequencing

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Q57: In Connection-Oriented transport, a connection is established:

Type: MCQ

- A) After data transmission
- B) During data transmission
- C) Before data transmission
- D) No connection is established

Correct Answer: C

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Q58: Connection Oriented transport is reliable and more secure.

Type: True / False

Correct Answer: True

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Q59: Describe the three-way handshake process in Connection-Oriented transport.

Type: Short Answer

Correct Answer: 1) Browser sends synchronization packet to web server. 2) Browser expects synchronization-acknowledgment response. 3) When it receives response, it sends its own response to the server. After this, data can be transmitted.

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Q60: Connectionless-Oriented transport is used with:

Type: MCQ

- A) File transfers
- B) Web browsing
- C) Delay-sensitive data like audio/video
- D) Email sending

Correct Answer: C

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Q61: Connectionless-Oriented transport establishes a connection before data transmission.

Type: True / False

Correct Answer: False

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Note: No connection establishment prior to data transmission.

Q62: Connectionless transport allows data flooding.

Type: True / False

Correct Answer: True

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Q63: Connectionless-Oriented is used with delay-sensitive data such as _____ and _____.

Type: Fill in the Blank

Correct Answer: Audio Calls / Video Streaming

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Q64: Which of the following is NOT a function of the Transport Layer?

Type: MCQ

- A) Session Multiplexing
- B) Segmentation
- C) Routing
- D) Flow Control

Correct Answer: C

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Note: Routing is a Network Layer function.

Q65: The Transport Layer adds _____ to identify services.

Type: Fill in the Blank

Correct Answer: Port Number

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Q66: The Transport Layer provides Socket to Socket communication.

Type: True / False

Correct Answer: True

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Q67: Port numbers range from:

Type: MCQ

- A) 0 to 1024
- B) 0 to 65,535
- C) 1 to 255
- D) 0 to 32,768

Correct Answer: B

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Q68: Port numbers are _____ bits in length.

Type: Fill in the Blank

Correct Answer: 16

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Q69: The well-known port for HTTP is:

Type: MCQ

- A) 21
- B) 25
- C) 53
- D) 80

Correct Answer: D

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Q70: The well-known port for FTP is:

Type: MCQ

- A) 80
- B) 21
- C) 53
- D) 25

Correct Answer: B

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Q71: The well-known port for DNS is:

Type: MCQ

- A) 25
- B) 80
- C) 53
- D) 21

Correct Answer: C

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Q72: The well-known port for SMTP is:

Type: MCQ

- A) 80
- B) 21
- C) 53
- D) 25

Correct Answer: D

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Q73: Well-known ports range from _____ to _____.

Type: Fill in the Blank

Correct Answer: 0 / 1024

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Q74: Ephemeral ports are ports greater than _____.

Type: Fill in the Blank

Correct Answer: 1024

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Q75: Match each protocol with its well-known port number:

Type: Matching

- 1) HTTP
- 2) FTP
- 3) DNS
- 4) SMTP

Match with:

- a) 21
- b) 25
- c) 53
- d) 80

Correct Answer: 1-d, 2-a, 3-c, 4-b

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Q76: UDP provides:

Type: MCQ

- A) Reliable delivery
- B) Best-effort delivery
- C) Connection-oriented delivery
- D) Sequenced delivery

Correct Answer: B

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Q77: UDP is a connectionless protocol.

Type: True / False

Correct Answer: True

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Q78: UDP has data-recovery features.

Type: True / False

Correct Answer: False

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Note: UDP has no data-recovery features.

Q79: UDP data loss can occur due to _____ error or _____.

Type: Fill in the Blank

Correct Answer: Routing / Duplication

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Q80: Why does UDP save overhead compared to TCP?

Type: Short Answer

Correct Answer: UDP saves the overhead of reliability mechanisms. It has limited error checking and no data-recovery features.

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Q81: TCP is a _____ protocol.

Type: MCQ

- A) Connectionless
- B) Best-effort
- C) Connection-oriented
- D) Unreliable

Correct Answer: C

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Q82: TCP provides sequencing of data packets.

Type: True / False

Correct Answer: True

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Q83: TCP provides acknowledgement of receipt.

Type: True / False

Correct Answer: True

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Q84: UDP is more commonly used than TCP.

Type: True / False

Correct Answer: False

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Note: TCP is more commonly used than UDP.

Q85: List all features of TCP mentioned in the lecture.

Type: Short Answer

Correct Answer: Reliable protocol, provides access to network layer, connection-oriented, error checking, data-recovery features, sequencing of data packets, acknowledgement of receipt, more commonly used than UDP.

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Q86: Match each characteristic with the correct protocol:

Type: Matching

- 1) Best-effort delivery
- 2) Reliable delivery

- 3) No data recovery
- 4) Acknowledgement of receipt

Match with:

- a) TCP
- b) UDP

Correct Answer: 1-b, 2-a, 3-b, 4-a

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SECTION 5: NETWORK LAYER

Pages 16-19

Q87: The Network Layer organizes data into:

Type: MCQ

- A) Segments
- B) Packets
- C) Frames
- D) Bits

Correct Answer: B

Page: Page 16

Q88: The Network Layer is responsible for end-to-end addressing and routing.

Type: True / False

Correct Answer: True

Page: Page 16

Q89: The Network Layer identifies a unique _____ address for machines.

Type: Fill in the Blank

Correct Answer: logical

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Q90: Which device works in the Network Layer?

Type: MCQ

- A) Hub
- B) Switch
- C) Router
- D) Repeater

Correct Answer: C

Page: Page 16

Q91: The Network Layer performs fragmentation if segmentation is not enough.

Type: True / False

Correct Answer: True

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Q92: What factors determine the route selection in the Network Layer?

Type: Short Answer

Correct Answer: Network overload, Quality of Service (QoS), Cost of alternative routes, Delivery priorities.

Q93: The Network Layer converts segments to packets by adding _____ and _____ addresses.

Type: Fill in the Blank

Correct Answer: Source / Destination

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Q94: A logical address is:

Type: MCQ

- A) Burned on the NIC
- B) Bound to geographical location
- C) Vendor oriented
- D) Cannot be changed

Correct Answer: B

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Q95: A logical address can be changed.

Type: True / False

Correct Answer: True

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Q96: A logical address is vendor oriented.

Type: True / False

Correct Answer: False

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Note: Logical addresses are NOT vendor oriented.

Q97: The common example of a logical address is the _____ address in TCP/IP.

Type: Fill in the Blank

Correct Answer: IP

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Q98: The Network Layer defines:

Type: MCQ

- A) Source and Destination Machine
- B) Hop-to-hop addressing
- C) Bit stream transmission
- D) Session management

Correct Answer: A

Page: Page 19

Q99: What does the Transport Layer guarantee according to the used protocol?

Type: Short Answer

Correct Answer: Either End-to-End Reliable Delivery or End-to-End Best-Effort Delivery.

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SECTION 6: DATA LINK LAYER

Pages 20-26

Q100: The Data Link Layer organizes data into:

Type: MCQ

- A) Packets
- B) Segments
- C) Frames
- D) Bits

Correct Answer: C

Page: Page 21

Q101: The Data Link Layer is responsible for physical addressing.

Type: True / False

Correct Answer: True

Page: Page 21

Q102: The Data Link Layer provides _____ to _____ addressing.

Type: Fill in the Blank

Correct Answer: Hop / hop

Page: Page 21

Q103: The Data Link Layer checks frames for:

Type: MCQ

- A) Routing paths
- B) Logical addresses
- C) Errors
- D) Port numbers

Correct Answer: C

Page: Page 21

Q104: The Data Link Layer puts frames on the physical medium.

Type: True / False

Correct Answer: True

Page: Page 21

Q105: A physical address (MAC) is:

Type: MCQ

- A) Bound to geographical location
- B) Can be changed

- C) Burned on the NIC
- D) Not vendor oriented

Correct Answer: C

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Q106: A MAC address can be changed.

Type: True / False

Correct Answer: False

Page: Page 22

Note: MAC address is burned on the NIC and cannot be changed.

Q107: A physical address is vendor oriented.

Type: True / False

Correct Answer: True

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Q108: A physical address is sometimes called _____ address.

Type: Fill in the Blank

Correct Answer: hardware

Page: Page 22

Q109: Match each property with the correct address type (Logical vs Physical):

Type: Matching

- 1) Bound to geographical location
- 2) Burned on NIC
- 3) Can be changed
- 4) Vendor oriented

Match with:

- a) Logical Address
- b) Physical Address

Correct Answer: 1-a, 2-b, 3-a, 4-b

Page: Pages 18, 22

Q110: FCS stands for:

Type: MCQ

- A) Frame Control System
- B) Frame Check Sequence
- C) File Check System
- D) Frame Coding Standard

Correct Answer: B

Page: Page 24

Q111: What happens when FCS comparison fails?

Type: Short Answer

Correct Answer: Data is assumed corrupted and the receiver requests data resend.

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Q112: If the receiver does not receive the success message during specified time, the sender automatically resends the data frames.

Type: True / False

Correct Answer: True

Page: Page 25

Q113: FCS number is based on the values of all _____ fields in the frame.

Type: Fill in the Blank

Correct Answer: preceding

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Q114: The Data Link Layer is divided into sublayers by:

Type: MCQ

- A) ISO
- B) IETF
- C) IEEE
- D) ITU

Correct Answer: C

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Q115: The top sublayer of the Data Link Layer is:

Type: MCQ

- A) MAC
- B) LLC
- C) FCS
- D) NIC

Correct Answer: B

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Q116: LLC stands for _____ .

Type: Fill in the Blank

Correct Answer: Logical Link Control

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Q117: MAC stands for _____ .

Type: Fill in the Blank

Correct Answer: Media Access Control

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Q118: What are the functions of the LLC sublayer?

Type: Short Answer

Correct Answer: Communicating only with Network layer, providing hop-to-hop reliability and hop-to-hop flow control.

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Q119: What is the MAC sublayer responsible for?

Type: Short Answer

Correct Answer: Adding physical address to the frame and communication with the Physical layer.

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Q120: Which devices work in the Data Link Layer?

Type: MCQ

- A) Routers and Hubs
- B) Switches and Bridges
- C) Repeaters and NIC
- D) Routers and Layer 3 Switches

Correct Answer: B

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Q121: If part of the frame is lost during transmission, the Data Link Layer at the receiver requests data retransmit from the sender.

Type: True / False

Correct Answer: True

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SECTION 7: PHYSICAL LAYER

Pages 27-28

Q122: The Physical Layer transmits:

Type: MCQ

- A) Frames
- B) Packets
- C) Segments
- D) Bit stream

Correct Answer: D

Page: Page 27

Q123: The Physical Layer is responsible for the representation of 1s and 0s.

Type: True / False

Correct Answer: True

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Q124: Copper cables include _____ and _____ types.

Type: Fill in the Blank

Correct Answer: Coaxial / Twisted Pair

Page: Page 27

Q125: Fiber Optic cables come in _____ Mode and _____ Mode.

Type: Fill in the Blank

Correct Answer: Single / Multi

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Q126: Which devices work in the Physical Layer?

Type: MCQ

- A) Router, Switch, Bridge
- B) NIC, Hub, Repeater
- C) Switch, Bridge, NIC
- D) Router, Hub, NIC

Correct Answer: B

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Q127: List all types of transmission media mentioned for the Physical Layer.

Type: Short Answer

Correct Answer: Cables: Copper (Coaxial, Twisted Pair), Fiber Optic (Single Mode, Multi Mode), and Wireless.

Q128: Wireless is a type of transmission media in the Physical Layer.

Type: True / False

Correct Answer: True

Q129: The Physical Layer adapts to the:

Type: MCQ

- A) Application software
- B) Transport protocol
- C) Transmission media
- D) Routing table

Correct Answer: C

SECTION 8: ISO/OSI MODEL

Pages 29-33

Q130: ISO/OSI is a:

Type: MCQ

- A) Physical device
- B) Conceptual model
- C) Protocol
- D) Software application

Correct Answer: B

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Q131: The goal of ISO/OSI is the _____ of diverse communication systems.

Type: Fill in the Blank

Correct Answer: interoperability

Page: Page 30

Q132: The original OSI model defined seven layers.

Type: True / False

Correct Answer: True

Page: Page 30

Q133: ISO/OSI partitions a communication system into _____ layers.

Type: Fill in the Blank

Correct Answer: abstraction

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Q134: Each OSI layer serves the layer above it and is served by the layer below it.

Type: True / False

Correct Answer: True

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Q135: What does OSI stand for?

Type: Short Answer

Correct Answer: Open System Interconnection.

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Q136: Two instances at the same OSI layer are visualized as connected by:

Type: MCQ

- A) A vertical connection
- B) A horizontal connection
- C) No connection
- D) A physical cable

Correct Answer: B

Page: Page 31

Q137: The OSI model works regardless of the underlying internal structure and technology.

Type: True / False

Correct Answer: True

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SECTION 9: TCP/IP PROTOCOL SUITE

Pages 39-45

Q138: A computer network protocol is:

Type: MCQ

- A) A physical device
- B) A set of rules and agreements governing communication
- C) A type of cable
- D) An operating system

Correct Answer: B

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Q139: TCP/IP is the standard protocol over the Internet.

Type: True / False

Correct Answer: True

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Q140: TCP/IP is vendor oriented.

Type: True / False

Correct Answer: False

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Note: TCP/IP is open for development and not vendor oriented.

Q141: TCP/IP is _____ compatible.

Type: Fill in the Blank

Correct Answer: ISO/OSI

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Q142: List the services offered by TCP/IP.

Type: Short Answer

Correct Answer: Web browsing, file transfer, and e-mail exchange.

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Q143: How many layers does TCP/IP have compared to OSI's 7?

Type: MCQ

- A) 3
- B) 4
- C) 5
- D) 6

Correct Answer: B

Q144: How are OSI layers merged in TCP/IP?

Type: Short Answer

Correct Answer: Physical and Data Link merge into Network Access layer. Network and Transport stay separate. The upper 3 layers (Application, Presentation, Session) merge into the Application layer.

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Q145: In TCP/IP, the physical and data link layers are merged into the _____ layer.

Type: Fill in the Blank

Correct Answer: Network Access (or Physical)

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Q146: TCP/IP is the default protocol for Microsoft, Linux, and many other operating systems.

Type: True / False

Correct Answer: True

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Q147: Which versions of TCP/IP are currently running?

Type: MCQ

- A) Version 1 and 2
- B) Version 3 and 5
- C) Version 4 and 6
- D) Version 5 and 7

Correct Answer: C

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Q148: Network protocols include NetBEUI, IPX/SPX, _____, and TCP/IP.

Type: Fill in the Blank

Correct Answer: Apple Talk

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SECTION 10: IP PROTOCOL & ADDRESSING

Pages 46-55

Q149: IP is the _____ communication protocol in the TCP/IP suite.

Type: MCQ

- A) Secondary
- B) Optional
- C) Principal
- D) Backup

Correct Answer: C

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Q150: IP delivers packets solely based on the IP addresses in the packet headers.

Type: True / False

Correct Answer: True

Page: Page 47

Q151: The IP protocol defines _____ to label datagrams with source and destination addresses.

Type: Fill in the Blank

Correct Answer: addressing

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Q152: An IPv4 address is how many bits long?

Type: MCQ

- A) 16 bits
- B) 32 bits
- C) 64 bits
- D) 128 bits

Correct Answer: B

Page: Page 48

Q153: An IPv4 address consists of _____ Bytes (or Octets).

Type: Fill in the Blank

Correct Answer: 4

Page: Page 48

Q154: Each byte in an IP address ranges from 0 to 255.

Type: True / False

Correct Answer: True

Q155: IP addresses are categorized into classes:

Type: MCQ

- A) A, B, C only
- B) A, B, C, D, E
- C) 1, 2, 3, 4, 5
- D) I, II, III, IV

Correct Answer: B

Page: Page 48

Q156: 127 is a Class A address reserved for loopback testing.

Type: True / False

Correct Answer: True

Page: Page 49

Q157: Address 127 can be assigned to a network.

Type: True / False

Correct Answer: False

Page: Page 49

Note: 127 is reserved for loopback testing and cannot be assigned to a network.

Q158: The private IP range for Class A is:

Type: MCQ

- A) 172.16.0.0 to 172.31.255.255
- B) 192.168.0.0 to 192.168.255.255
- C) 10.0.0.0 to 10.255.255.255
- D) 127.0.0.0 to 127.255.255.255

Correct Answer: C

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Q159: The private IP range for Class B is:

Type: MCQ

- A) 10.0.0.0 to 10.255.255.255
- B) 172.16.0.0 to 172.31.255.255
- C) 192.168.0.0 to 192.168.255.255
- D) 169.254.0.0 to 169.254.255.255

Correct Answer: B

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Q160: The private IP range for Class C is:

Type: MCQ

- A) 10.0.0.0 to 10.255.255.255

- B) 172.16.0.0 to 172.31.255.255
- C) 192.168.0.0 to 192.168.255.255
- D) 224.0.0.0 to 239.255.255.255

Correct Answer: C

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Q161: Match each class with its private IP range:

Type: Matching

- 1) Class A
- 2) Class B
- 3) Class C

Match with:

- a) 192.168.0.0 to 192.168.255.255
- b) 10.0.0.0 to 10.255.255.255
- c) 172.16.0.0 to 172.31.255.255

Correct Answer: 1-b, 2-c, 3-a

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Q162: Which devices work in the IP layer?

Type: MCQ

- A) Hub and Repeater
- B) Switch and Bridge
- C) Router and Layer 3 Switch
- D) NIC and Hub

Correct Answer: C

Page: Page 53

Q163: Network ID = IP Address AND _____.

Type: Fill in the Blank

Correct Answer: Subnet mask

Page: Page 53

Q164: Layer 3 switches are super fast routers.

Type: True / False

Correct Answer: True

Page: Page 53

Q165: Routers are protocol dependent.

Type: True / False

Correct Answer: True

Page: Page 53

Q166: List the components of a Routing Table.

Type: Short Answer

Correct Answer: Destination Network, Subnet Mask, Interface, Gateway, Cost/Metric.

Page: Page 54

Q167: A routing table can be built statically or dynamically.

Type: True / False

Correct Answer: True

Page: Page 54

Q168: _____ and _____ are dynamic routing protocols.

Type: Fill in the Blank

Correct Answer: RIP / OSPF

Page: Page 54

Q169: Routing table rows are read:

Type: MCQ

- A) Randomly
- B) In parallel
- C) Sequentially
- D) In reverse

Correct Answer: C

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SECTION 11: APPLICATION PROTOCOLS

Pages 66-75

Q170: HTTP uses port number:

Type: MCQ

- A) 21
- B) 443
- C) 80
- D) 25

Correct Answer: C

Page: Page 68

Q171: HTTPS uses port number:

Type: MCQ

- A) 80
- B) 8080
- C) 443
- D) 22

Correct Answer: C

Page: Page 68

Q172: HTTP transfers text, images, and hyper links.

Type: True / False

Correct Answer: True

Page: Page 68

Q173: The secure version of HTTP is _____.

Type: Fill in the Blank

Correct Answer: HTTPS

Page: Page 68

Q174: FTP uses port number:

Type: MCQ

- A) 80
- B) 25
- C) 53
- D) 21

Correct Answer: D

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Q175: FTP is built-in in MS Windows Operating Systems.

Type: True / False

Correct Answer: True

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Q176: A web browser can work as an FTP client.

Type: True / False

Correct Answer: True

Page: Page 69

Q177: Which protocol collects email messages from the mail server?

Type: MCQ

A) SMTP

B) IMAP4

C) POP3

D) HTTP

Correct Answer: C

Page: Page 71

Q178: What is the difference between POP3 and IMAP4?

Type: Short Answer

Correct Answer: POP3 downloads full messages to the machine (user can work offline, saves server quota). IMAP4 retrieves the message header only.

Page: Page 71

Q179: SMTP is the common language used by the majority of Mail Servers.

Type: True / False

Correct Answer: True

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Q180: DNS resolves _____ into _____.

Type: Fill in the Blank

Correct Answer: Domain Name / IP Address

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Q181: DNS stands for:

Type: MCQ

A) Data Name Service

B) Domain Name System

C) Domain Network Service

D) Data Network System

Correct Answer: B

Page: Page 72

Q182: DHCP stands for _____ .

Type: Fill in the Blank

Correct Answer: Dynamic Host Configuration Protocol

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SECTION 12: TCP/IP TOOLS

Pages 76-83

Q183: What is the IPCONFIG tool used for?

Type: Short Answer

Correct Answer: Used to view and manage IP configuration settings.

Page: Page 77

Q184: The PING command option '-t' means:

Type: MCQ

- A) Ping a specific number of times
- B) Ping continuously
- C) Record route
- D) Set timeout

Correct Answer: B

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Q185: Match each PING option with its function:

Type: Matching

- 1) -t
- 2) -n
- 3) -r
- 4) -w

Match with:

- a) Record route
- b) Set timeout/wait
- c) Continuous ping
- d) Number of pings

Correct Answer: 1-c, 2-d, 3-a, 4-b

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Q186: What does the ARP tool do?

Type: Short Answer

Correct Answer: ARP resolves IP addresses to MAC (physical) addresses.

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Q187: Match each FTP command with its function:

Type: Matching

- 1) open
- 2) ls
- 3) cd

- 4) get
- 5) put
- 6) bye

Match with:

- a) List files
- b) Connect to server
- c) Change directory
- d) Download file
- e) Upload file
- f) Exit FTP

Correct Answer: 1-b, 2-a, 3-c, 4-d, 5-e, 6-f

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Q188: The _____ tool is used for DNS queries.

Type: Fill in the Blank

Correct Answer: NSLOOKUP

Page: Page 81

Q189: Which of the following is NOT a Route command?

Type: MCQ

- A) Route Print
- B) Route Add
- C) Route Remove
- D) Route Delete

Correct Answer: C

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Note: The correct commands are Route Print, Add, Delete, and Change.

Q190: List all Route commands mentioned in the lecture.

Type: Short Answer

Correct Answer: Route Print, Route Add, Route Delete, Route Change.

Page: Page 82

Q191: Match each NETSTAT option with its function:

Type: Matching

- 1) -a
- 2) -n
- 3) -p

Match with:

- a) Show addresses numerically
- b) Show all connections
- c) Show per-protocol statistics

Correct Answer: 1-b, 2-a, 3-c

Page: Page 83

Q192: Netstat -? shows the _____ for the netstat command.

Type: Fill in the Blank

Correct Answer: help

Page: Page 83

SECTION 13: REVIEW QUESTIONS FROM LECTURE

Pages 34-38

Q193: The computer network is composed of computers, _____, and _____ connected through transmission media.

Type: Fill in the Blank

Correct Answer: Peripherals / Connecting Devices

Page: Page 34

Q194: Transmission media can either be _____ made of _____ or _____, or it can be _____ using the _____ as a media.

Type: Fill in the Blank

Correct Answer: Wired / Copper / Fiber Optic / Wireless / Air

Page: Page 34

Q195: The most commonly used network topology nowadays is the _____ topology, connected to a central device which is the _____.

Type: Fill in the Blank

Correct Answer: Star / Switch

Page: Page 35

Q196: Based on geographical area, the network can be a _____ or a _____.

Type: Fill in the Blank

Correct Answer: LAN / WAN

Page: Page 35

Q197: According to administration type, a network can be either _____ or _____.

Type: Fill in the Blank

Correct Answer: Peer-to-peer / Client/Server

Page: Page 35

Q198: The most commonly used copper cable is _____ divided into _____ and _____.

Type: Fill in the Blank

Correct Answer: Twisted Pair / STP / UTP

Page: Page 36

Q199: Fiber optic advantages over copper: it is _____, _____, and _____.

Type: Fill in the Blank

Correct Answer: Faster / More immune to EMI / More secure

Q200: OSI stands for _____.

Type: Fill in the Blank

Correct Answer: Open System Interconnection

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Q201: The most commonly used protocol over the Internet is _____.

Type: Fill in the Blank

Correct Answer: TCP/IP

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Q202: The transport layer can work in either _____ or _____ modes of delivery.

Type: Fill in the Blank

Correct Answer: Reliable / Best-effort

Page: Page 38

Q203: The network layer is responsible of _____ addressing, while the data link layer is responsible of _____ addressing.

Type: Fill in the Blank

Correct Answer: Logical / Physical

Page: Page 38

Q204: The main functionalities of the network layer are the _____ and _____.

Type: Fill in the Blank

Correct Answer: Addressing / Routing

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FINAL EXAM — HARD MODE

50 Advanced Questions — Combined Concepts & Application-Based

Q205: Which layer is responsible for BOTH data encryption AND data compression?

Type: MCQ

- A) Application
- B) Session
- C) Presentation
- D) Transport

Correct Answer: C

Page: Page 4

Q206: If a segment arrives out of order, which Transport Layer function handles this?

Type: MCQ

- A) Segmentation
- B) Sequencing
- C) Flow control
- D) Deduplication

Correct Answer: B

Page: Pages 12-13

Q207: A router searches the Routing Table for the destination using:

Type: MCQ

- A) MAC address AND subnet mask
- B) IP address AND subnet mask
- C) Port number AND IP address
- D) MAC address AND port number

Correct Answer: B

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Q208: Which of the following correctly maps OSI layers to TCP/IP layers?

Type: MCQ

- A) OSI 1-2 → Transport, OSI 3-4 → Network, OSI 5-7 → Application
- B) OSI 1-2 → Network Access, OSI 3 → Network, OSI 4 → Transport, OSI 5-7 → Application
- C) OSI 1-3 → Network, OSI 4-5 → Transport, OSI 6-7 → Application
- D) OSI 1 → Physical, OSI 2-4 → Network, OSI 5-7 → Application

Correct Answer: B

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Q209: A user is streaming live video. Which transport protocol is being used and why?

Type: MCQ

- A) TCP - because it is reliable
- B) UDP - because it is used with delay-sensitive real-time data
- C) TCP - because it has error recovery
- D) UDP - because it establishes connection first

Correct Answer: B

Page: Pages 15, 61

Q210: The FCS is created at which layer and what happens on comparison failure?

Type: MCQ

- A) Network Layer — packet is dropped
- B) Transport Layer — segment is resent
- C) Data Link Layer — data is assumed corrupted and resend is requested
- D) Physical Layer — bits are retransmitted

Correct Answer: C

Page: Page 24

Q211: Which sublayer adds the physical address to the frame?

Type: MCQ

- A) LLC
- B) MAC
- C) FCS
- D) NIC

Correct Answer: B

Page: Page 26

Q212: What is the key difference between logical and physical addresses regarding vendor orientation?

Type: MCQ

- A) Both are vendor oriented
- B) Logical is vendor oriented, Physical is not
- C) Logical is NOT vendor oriented, Physical IS vendor oriented
- D) Neither is vendor oriented

Correct Answer: C

Page: Pages 18, 22

Q213: In the OSI model, the Session Layer and the Transport Layer BOTH handle flow control.

Type: True / False

Correct Answer: False

Page: Pages 7, 10

Note: Flow control is handled by the Transport Layer, not the Session Layer.

Q214: TCP provides both error checking and data-recovery features, while UDP provides neither.

Type: True / False

Correct Answer: False

Page: Pages 61, 63

Note: UDP does have limited error checking, but no data-recovery features.

Q215: An IP address 10.5.3.1 belongs to which class and is it public or private?

Type: MCQ

- A) Class A, Public
- B) Class A, Private
- C) Class B, Private
- D) Class C, Public

Correct Answer: B

Page: Pages 48-51

Q216: An IP address 192.168.1.100 belongs to which class and type?

Type: MCQ

- A) Class B, Private
- B) Class C, Public
- C) Class C, Private
- D) Class A, Private

Correct Answer: C

Page: Pages 48-51

Q217: Which layer converts segments into packets by adding addressing information?

Type: MCQ

- A) Transport
- B) Data Link
- C) Network
- D) Session

Correct Answer: C

Page: Page 17

Q218: In the three-way handshake, the correct order is:

Type: MCQ

- A) ACK → SYN → SYN-ACK
- B) SYN → SYN-ACK → ACK
- C) SYN-ACK → SYN → ACK
- D) ACK → SYN-ACK → SYN

Correct Answer: B

Page: Page 14

Q219: IMAP4 downloads the complete email message to the local machine just like POP3.

Type: True / False

Correct Answer: False

Q220: If a sender does not receive a 'success' message during the specified time, what happens?

Type: MCQ

- A) Connection is terminated
- B) Sender automatically resends data frames
- C) Receiver resends the success message
- D) Data is dropped

Correct Answer: B

Page: Page 25

Q221: The routing function of IP essentially establishes:

Type: MCQ

- A) The LAN
- B) The WAN
- C) The Internet
- D) The Intranet

Correct Answer: C

Page: Page 47

Q222: Which of the following is TRUE about the Session Layer?

Type: MCQ

- A) It handles data encryption
- B) It defines node communication order and timing
- C) It organizes data into segments
- D) It provides physical addressing

Correct Answer: B

Page: Page 7

Q223: A socket address combines the _____ address and the _____ number.

Type: Fill in the Blank

Correct Answer: IP / Port

Page: Pages 57-59

Q224: Which tool resolves IP addresses to MAC addresses?

Type: MCQ

- A) NSLOOKUP
- B) PING
- C) ARP
- D) NETSTAT

Correct Answer: C

Page: Page 79

Q225: Which tool is used for DNS queries?

Type: MCQ

- A) IPCONFIG
- B) NSLOOKUP
- C) ROUTE
- D) ARP

Correct Answer: B

Page: Page 81

Q226: A physical (MAC) address is bound to geographical location.

Type: True / False

Correct Answer: False

Page: Page 22

Note: Physical addresses are NOT bound to geographical location; logical addresses are.

Q227: Data deduplication and loss recovery are functions of which layer?

Type: MCQ

- A) Network
- B) Data Link
- C) Physical
- D) Transport

Correct Answer: D

Page: Page 10

Q228: The LLC sublayer communicates with which layer?

Type: MCQ

- A) Physical
- B) Transport
- C) Network
- D) Session

Correct Answer: C

Page: Page 26

Q229: RIP and OSPF are static routing protocols.

Type: True / False

Correct Answer: False

Page: Page 54

Note: RIP and OSPF are dynamic routing protocols.

Q230: Which of the following CANNOT be changed?

Type: MCQ

- A) IP address

- B) Logical address
- C) MAC address
- D) Port number

Correct Answer: C

Page: Page 22

Q231: The correct data unit transformation through layers (top to bottom) is:

Type: MCQ

- A) Frames → Packets → Segments → Bits
- B) Data → Segments → Packets → Frames → Bits
- C) Bits → Frames → Packets → Segments
- D) Segments → Bits → Frames → Packets

Correct Answer: B

Page: Pages 10, 16, 21, 27

Q232: Which address type is used for end-to-end routing?

Type: MCQ

- A) Physical address
- B) MAC address
- C) Logical (IP) address
- D) Hardware address

Correct Answer: C

Page: Page 18

Q233: Which address type is used for hop-to-hop addressing?

Type: MCQ

- A) IP address
- B) Logical address
- C) MAC (Physical) address
- D) Port number

Correct Answer: C

Page: Pages 21-22

Q234: The Network Layer selects the best path for destination.

Type: True / False

Correct Answer: True

Page: Page 16

Q235: An ephemeral port number could be:

Type: MCQ

- A) 80
- B) 21
- C) 1025
- D) 53

Correct Answer: C

Page: Page 58

Note: Ephemeral ports are greater than 1024.

Q236: Quality of Service (QoS) affects which layer's decision-making?

Type: MCQ

- A) Application
- B) Transport
- C) Network
- D) Physical

Correct Answer: C

Page: Page 17

Q237: The Data Link Layer provides end-to-end addressing.

Type: True / False

Correct Answer: False

Page: Page 21

Note: Data Link Layer provides hop-to-hop addressing.

Q238: Which combination correctly describes TCP?

Type: MCQ

- A) Connectionless, unreliable, no sequencing
- B) Connection-oriented, reliable, sequencing, acknowledgement
- C) Connectionless, reliable, no acknowledgement
- D) Connection-oriented, unreliable, no recovery

Correct Answer: B

Page: Page 63

Q239: SMTP is used to:

Type: MCQ

- A) Download emails
- B) Retrieve email headers
- C) Send messages between mail servers
- D) Resolve domain names

Correct Answer: C

Page: Page 71

Q240: POP3 saves quota on the server because it downloads messages to the machine.

Type: True / False

Correct Answer: True

Page: Page 71

Q241: If Network overload occurs, which layer adapts the routing path?

Type: MCQ

- A) Transport
- B) Session
- C) Network
- D) Data Link

Correct Answer: C

Page: Page 17

Q242: The FTP command 'mget' is used for:

Type: MCQ

- A) Upload single file
- B) Download multiple files
- C) List files
- D) Exit FTP

Correct Answer: B

Page: Page 80

Q243: Which of the following is NOT a component of the Routing Table?

Type: MCQ

- A) Destination Network
- B) Subnet Mask
- C) MAC Address
- D) Gateway

Correct Answer: C

Page: Page 54

Q244: The Physical Layer sends data signals to media and receives them.

Type: True / False

Correct Answer: True

Page: Page 27

Q245: 172.20.5.1 is a _____ IP address of Class _____.

Type: MCQ

- A) Public, A
- B) Private, B
- C) Public, B
- D) Private, C

Correct Answer: B

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Q246: Which layer handles BOTH session multiplexing AND segmentation?

Type: MCQ

- A) Network
- B) Transport

- C) Session
- D) Data Link

Correct Answer: B

Page: Page 57

Q247: DNS resolves IP addresses into domain names.

Type: True / False

Correct Answer: False

Page: Page 72

Note: DNS resolves Domain Names into IP Addresses.

Q248: In the TCP/IP model, which layers remain separate just like in the OSI model?

Type: MCQ

- A) Application and Presentation
- B) Network and Transport
- C) Data Link and Physical
- D) Session and Application

Correct Answer: B

Page: Page 44

Q249: The correct order of OSI layers from bottom to top is:

Type: MCQ

- A) Physical, Data Link, Network, Transport, Session, Presentation, Application
- B) Application, Presentation, Session, Transport, Network, Data Link, Physical
- C) Physical, Network, Data Link, Transport, Session, Application, Presentation
- D) Physical, Data Link, Transport, Network, Session, Presentation, Application

Correct Answer: A

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Q250: A protocol that includes addressing, routing, session management, and error recovery is called a:

Type: MCQ

- A) Network topology
- B) Computer network protocol
- C) Transmission media
- D) Physical device

Correct Answer: B

Page: Page 40

Q251: Which protocol is NOT associated with email services?

Type: MCQ

- A) SMTP
- B) POP3
- C) IMAP4

D) TFTP

Correct Answer: D

Page: Pages 67, 71

Q252: MTU acknowledgment happens before establishing connection with the recipient.

Type: True / False

Correct Answer: False

Page: Page 11

Note: MTU acknowledgment is during discovering routine and while establishing connection.

Q253: Netstat -a shows:

Type: MCQ

- A) Routing table
- B) All connections
- C) DNS cache
- D) IP configuration

Correct Answer: B

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COMPLETE ANSWER KEY TABLE

Fast Revision Reference — All Questions

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Q4	False	Page 2
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Q6	User interface to lower layers, Preparation of data for e...	Page 2
Q7	C	Page 3
Q8	B	Page 3
Q9	B	Page 3
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Q11	True	Page 3
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Q85	Reliable protocol, provides access to network layer, conn...	Page 63
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Q164	True	Page 53
Q165	True	Page 53
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Q167	True	Page 54
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Q172	True	Page 68
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Q175	True	Page 69
Q176	True	Page 69
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Q178	POP3 downloads full messages to the machine (user can wor...	Page 71
Q179	True	Page 71
Q180	Domain Name / IP Address	Page 72
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